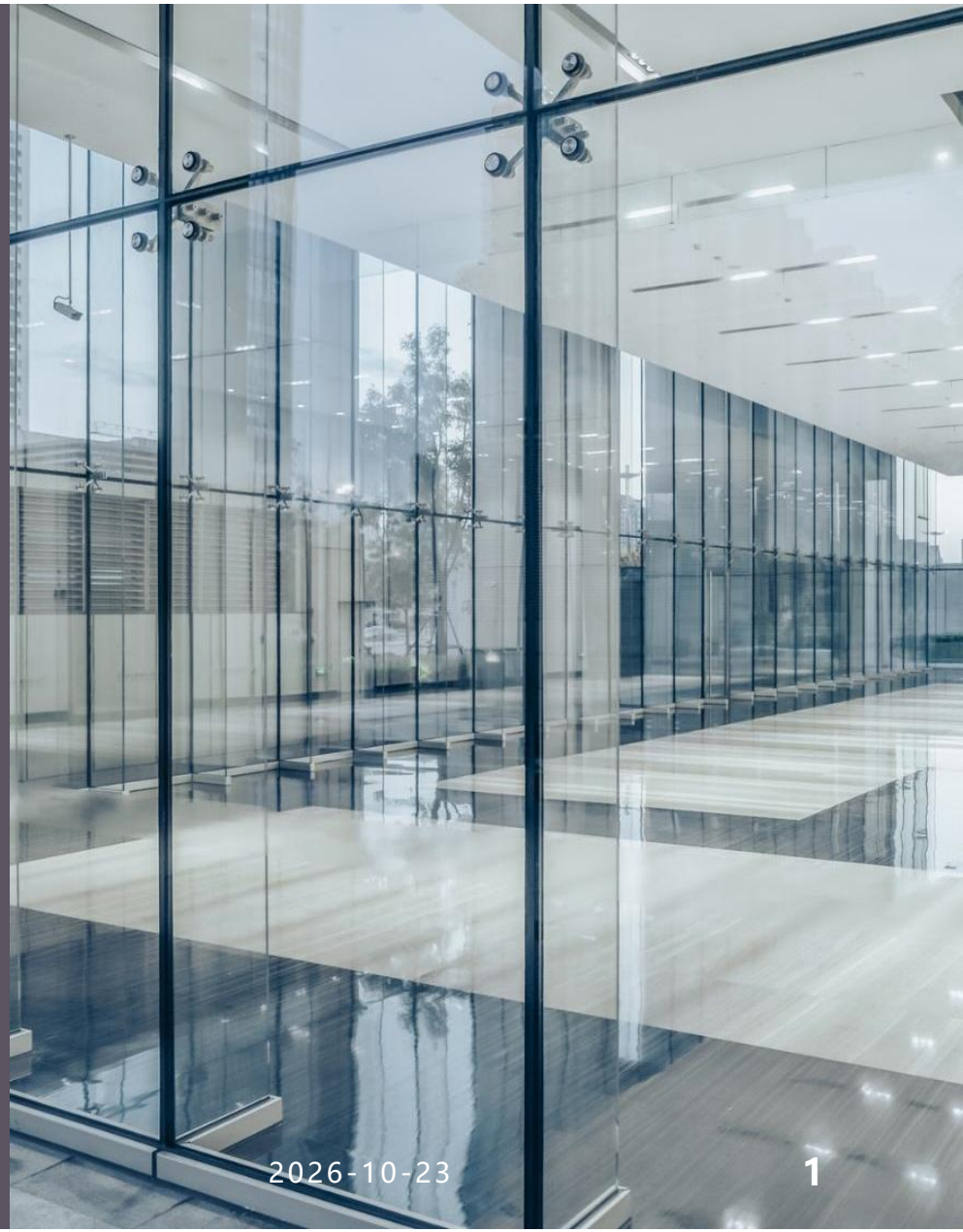
金融风险管理3.3

信用在险价值



西北农林科技大学 陈妍

信用VaR

信用VaR是某一展望期内，在一定置信度下，信用损失不会超过的数量。

有些信用VaR模型仅考虑由违约造成的损失，有些不仅考虑违约，还考虑由于降级和信用价差变化造成的损失。

当持有的产品是出于非交易目的时，信用VaR的展望期通常为1年。



目录

一、信用评级迁移矩阵

二、Vasicek模型

三、Credit Risk Plus

四、CreditMetrics

一、信用评级迁移矩阵



信用评级迁移矩阵

信用评级迁移矩阵揭示了在某一段时间内，公司的信用等级从一个级别迁移到另一个级别的概率。

这些概率基于历史数据得出，可能是内部评级生成的，也可能是专业机构评级发布的。



初始信用评级	一年期末的信用评级								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca-C	Default
Aaa	90.94%	8.36%	0.59%	0.08%	0.02%	0.00%	0.00%	0.00%	0.00%
Aa	0.87%	89.68%	8.84%	0.45%	0.07%	0.04%	0.02%	0.00%	0.02%
A	0.06%	2.64%	90.90%	5.67%	0.51%	0.12%	0.04%	0.01%	0.06%
Baa	0.04%	0.16%	4.44%	90.16%	4.09%	0.75%	0.17%	0.02%	0.18%
Ba	0.01%	0.05%	0.47%	6.66%	83.03%	7.90%	0.78%	0.12%	0.99%
B	0.01%	0.03%	0.16%	0.51%	5.32%	82.18%	7.39%	0.61%	3.79%
Caa	0.00%	0.01%	0.03%	0.11%	0.46%	7.82%	78.52%	3.30%	9.75%
Ca-C	0.00%	0.00%	0.07%	0.00%	0.80%	3.19%	11.41%	51.28%	33.24%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

一年期信用评级矩阵示例（穆迪，1970-2016）

➤ **TO: Final rating position** on 31-December-2025

		Investment grade										Speculative grade											
		Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca-C	WR	DEF
Investment grade	Aaa	**57.1**	36.7	0.0	0.0	4.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0
	Aa1	0.0	**72.7**	25.0	0.0	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Aa2	0.0	0.0	**94.6**	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.4	0.0
	Aa3	0.0	0.0	3.4	**93.1**	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.1	0.0
	A1	0.0	0.0	0.0	2.2	**91.4**	1.3	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0
	A2	0.0	0.0	0.0	0.0	7.1	**88.0**	1.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.4	0.0
	A3	0.0	0.0	0.0	0.0	0.0	6.6	**83.5**	4.5	0.4	0.6	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.3	0.0
	Baa1	0.0	0.0	0.0	0.0	0.2	0.0	8.5	**84.1**	3.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.9	0.0
	Baa2	0.0	0.0	0.0	0.0	0.0	0.2	0.3	7.1	**82.9**	3.8	0.5	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.1	0.0
	Baa3	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.4	13.0	**74.7**	2.6	1.5	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.9	0.0
Speculative grade	Ba1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3	13.0	**74.9**	2.7	1.3	0.4	0.0	0.0	0.0	0.0	0.0	0.0	6.3	0.0
	Ba2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.9	8.7	**77.9**	3.5	0.4	1.3	0.4	0.0	0.0	0.4	0.0	6.5	0.0
	Ba3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.2	15.4	**66.2**	6.9	1.5	0.4	0.0	0.0	0.0	0.4	8.1	0.0
	B1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	2.7	12.9	**63.5**	3.0	2.3	1.5	0.4	0.0	0.0	12.9	0.0
	B2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.4	0.9	11.9	**66.5**	7.0	0.9	1.8	0.0	0.4	9.3	0.4
	B3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	2.4	1.4	4.8	**59.0**	8.5	2.4	1.0	1.0	17.4	1.4
	Caa1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.0	7.1	**63.6**	10.8	3.2	0.5	12.7	1.6
	Caa2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	1.0	6.1	**65.1**	10.8	1.5	12.6	2.3
	Caa3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.7	0.4	2.1	5.0	**49.4**	15.1	15.5	10.9
	Ca-C	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	1.7	5.8	**40.1**	13.4	38.4

➤ **FROM: Initial rating position** on 01-January-2025

Source: Moody's Ratings

Average one-year alphanumeric rating migration rates, 1983-2025

Source: Moody's Ratings

From\To	Aaa	Aa1	Aa2	Aa3	A1	A2	A3	Baa1	Baa2	Baa3	Ba1	Ba2	Ba3	B1	B2	B3	Caa1	Caa2	Caa3	Ca_C	WR	Def
Aaa	87.15%	5.38%	2.09%	0.51%	0.33%	0.13%	0.02%	0.04%	0.00%	0.02%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.29%	0.00%
Aa1	1.56%	77.35%	7.81%	5.38%	1.38%	0.97%	0.19%	0.14%	0.07%	0.01%	0.03%	0.00%	0.01%	0.03%	0.02%	0.01%	0.02%	0.02%	0.00%	0.00%	4.99%	0.00%
Aa2	0.88%	3.98%	75.91%	9.27%	3.08%	1.44%	0.39%	0.08%	0.14%	0.06%	0.02%	0.01%	0.00%	0.02%	0.01%	0.02%	0.00%	0.02%	0.00%	0.00%	4.68%	0.00%
Aa3	0.13%	0.95%	4.18%	76.19%	8.82%	3.16%	0.77%	0.22%	0.23%	0.11%	0.02%	0.03%	0.01%	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	5.12%	0.04%
A1	0.05%	0.08%	0.94%	5.01%	78.21%	6.90%	2.30%	0.53%	0.35%	0.16%	0.14%	0.09%	0.04%	0.05%	0.01%	0.01%	0.01%	0.01%	0.00%	0.00%	5.06%	0.05%
A2	0.05%	0.03%	0.19%	0.88%	5.80%	77.95%	6.79%	2.22%	0.84%	0.31%	0.14%	0.11%	0.14%	0.04%	0.03%	0.01%	0.01%	0.02%	0.00%	0.00%	4.40%	0.04%
A3	0.03%	0.04%	0.14%	0.24%	1.23%	6.02%	77.46%	6.40%	2.20%	0.73%	0.29%	0.11%	0.10%	0.08%	0.03%	0.02%	0.02%	0.01%	0.00%	0.01%	4.78%	0.06%
Baa1	0.01%	0.02%	0.05%	0.09%	0.18%	1.21%	6.25%	78.15%	6.15%	1.83%	0.45%	0.23%	0.18%	0.20%	0.05%	0.02%	0.03%	0.02%	0.00%	0.02%	4.79%	0.09%
Baa2	0.03%	0.03%	0.01%	0.05%	0.13%	0.47%	1.53%	6.59%	77.67%	5.69%	1.09%	0.51%	0.34%	0.25%	0.15%	0.07%	0.08%	0.01%	0.01%	0.01%	5.12%	0.15%
Baa3	0.02%	0.00%	0.02%	0.03%	0.05%	0.14%	0.37%	1.53%	8.74%	74.36%	4.33%	1.80%	0.82%	0.64%	0.22%	0.21%	0.11%	0.06%	0.05%	0.03%	6.23%	0.24%
Ba1	0.01%	0.00%	0.01%	0.01%	0.12%	0.10%	0.19%	0.60%	2.14%	10.05%	66.27%	5.48%	3.67%	1.39%	0.53%	0.47%	0.14%	0.19%	0.03%	0.10%	8.01%	0.46%
Ba2	0.00%	0.00%	0.01%	0.02%	0.05%	0.09%	0.13%	0.29%	0.59%	3.41%	8.31%	65.65%	6.04%	3.46%	1.19%	0.77%	0.30%	0.20%	0.08%	0.11%	8.61%	0.66%
Ba3	0.00%	0.01%	0.01%	0.01%	0.05%	0.13%	0.14%	0.07%	0.36%	0.74%	2.71%	7.42%	65.25%	6.80%	2.85%	1.73%	0.67%	0.37%	0.09%	0.11%	9.26%	1.19%
B1	0.01%	0.01%	0.01%	0.01%	0.03%	0.02%	0.06%	0.07%	0.16%	0.35%	0.63%	2.74%	7.27%	63.73%	6.28%	4.24%	1.31%	0.71%	0.23%	0.23%	10.20%	1.69%
B2	0.00%	0.01%	0.00%	0.01%	0.01%	0.02%	0.08%	0.10%	0.12%	0.21%	0.18%	0.64%	2.06%	7.82%	62.49%	8.01%	3.43%	1.79%	0.42%	0.43%	9.50%	2.67%
B3	0.01%	0.00%	0.02%	0.00%	0.02%	0.02%	0.05%	0.03%	0.03%	0.08%	0.12%	0.25%	0.86%	2.61%	6.56%	60.51%	7.56%	3.39%	1.05%	0.76%	11.98%	4.07%
Caa1	0.00%	0.01%	0.00%	0.00%	0.00%	0.01%	0.00%	0.01%	0.01%	0.03%	0.04%	0.10%	0.22%	0.58%	1.19%	7.47%	59.81%	9.59%	2.76%	1.21%	13.33%	3.63%
Caa2	0.00%	0.00%	0.01%	0.00%	0.01%	0.01%	0.00%	0.00%	0.02%	0.04%	0.02%	0.02%	0.08%	0.23%	0.76%	1.55%	6.35%	61.56%	7.03%	2.66%	13.57%	6.06%
Caa3	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.03%	0.02%	0.02%	0.11%	0.28%	0.76%	2.81%	9.36%	47.93%	9.83%	13.44%	15.42%
Ca_C	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.02%	0.00%	0.00%	0.05%	0.10%	0.09%	0.11%	0.06%	0.33%	1.25%	1.42%	3.29%	6.20%	38.81%	14.80%	33.47%

信用评级迁移矩阵

假设在一定时间段内信用评级的变化独立于其他时间段，则可以利用1年期信用评级迁移矩阵，计算多年期的信用评级矩阵。

例如：2年期矩阵可以将两个1年期矩阵相乘；5年期矩阵是1年期矩阵的5次方。

马尔可夫矩阵

马尔可夫矩阵，也称转移概率矩阵，是一种特殊的方阵，用于描述系统从一个状态转移到另一个状态的概率。

马尔可夫矩阵的定义是基于马尔可夫性质，即**系统的未来状态只依赖于当前状态，而与过去的状态无关。**

马尔可夫矩阵的主要性质包括：

1、所有元素非负（$0 \le P_{ij} \le 1$）。
2、每一行的元素之和为1。
3、至少有一个特征值等于1，其他特征值的绝对值小于1。

马尔可夫矩阵可以用来预测系统状态的长期行为。通过计算马尔可夫矩阵的n次方，可以预测n步之后系统的状态分布。

初始信用评级	5年后的信用评级								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca-C	Default
Aaa	62.75%	28.14%	7.58%	1.16%	0.20%	0.07%	0.02%	0.00%	0.03%
Aa	2.96%	60.27%	29.98%	5.27%	0.82%	0.33%	0.13%	0.01%	0.20%
A	0.40%	9.00%	65.74%	19.73%	3.19%	1.08%	0.32%	0.04%	0.55%
Baa	0.18%	1.45%	15.45%	63.40%	12.25%	4.33%	1.18%	0.13%	1.68%
Ba	0.06%	0.39%	3.57%	19.72%	43.71%	19.78%	5.10%	0.55%	7.16%
B	0.04%	0.16%	0.94%	3.83%	13.22%	43.21%	16.56%	1.65%	20.39%
Caa	0.01%	0.06%	0.26%	0.94%	3.46%	17.61%	34.25%	3.53%	39.88%
Ca-C	0.00%	0.03%	0.21%	0.51%	2.09%	7.30%	12.50%	4.53%	72.82%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

根据表1计算出的五年期信用评级迁移矩阵

Average five-year letter rating migration rates, 1970-2025

Source: Moody's Ratings

From\To	Aaa	Aa	A	Baa	Ba	B	Caa	Ca_C	WR	Def
Aaa	54.79%	22.13%	4.61%	0.57%	0.28%	0.03%	0.04%	0.00%	17.48%	0.07%
Aa	1.89%	47.84%	23.47%	3.56%	0.69%	0.22%	0.10%	0.02%	21.98%	0.23%
A	0.15%	7.11%	54.59%	13.58%	2.02%	0.61%	0.12%	0.01%	21.25%	0.56%
Baa	0.11%	0.76%	10.99%	53.14%	7.09%	2.17%	0.45%	0.08%	24.02%	1.20%
Ba	0.02%	0.15%	2.01%	14.24%	29.77%	10.81%	2.02%	0.13%	34.85%	6.00%
B	0.02%	0.05%	0.35%	2.10%	8.03%	24.04%	6.82%	0.55%	43.21%	14.82%
Caa	0.00%	0.01%	0.06%	0.39%	1.31%	6.98%	17.75%	1.45%	50.89%	21.16%
Ca_C	0.00%	0.00%	0.00%	0.62%	1.24%	3.59%	4.22%	2.43%	37.57%	50.34%

信用评级迁移矩阵

计算6个月期信用评级迁移矩阵要将一年期信用评级迁移矩阵开方。

计算3个月期信用评级迁移矩阵要将一年期信用评级迁移矩阵开4次方。

计算1个月期信用评级迁移矩阵要将一年期信用评级迁移矩阵开12次方。（注意矩阵开方不是对矩阵中的每个数字分别开方）

矩阵开方

矩阵开方（Matrix Square Root）是指给定矩阵 A，求矩阵 X 使得 $X^2 = A$。与标量不同，矩阵平方根不一定唯一，甚至可能不存在。

存在性判断
正定/半正定矩阵：存在唯一的主平方根（半正定）
有负特征值：无实平方根，只有复平方根
可对角化 + 无零特征值：存在 2^n 个平方根

初始信用评级	一个月末的信用评级								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca-C	Default
Aaa	99.21%	0.77%	0.02%	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%
Aa	0.08%	99.08%	0.81%	0.02%	0.00%	0.00%	0.00%	0.00%	0.00%
A	0.00%	0.24%	99.19%	0.52%	0.04%	0.01%	0.00%	0.00%	0.00%
Baa	0.00%	0.01%	0.40%	99.12%	0.39%	0.05%	0.01%	0.00%	0.01%
Ba	0.00%	0.00%	0.03%	0.63%	98.42%	0.78%	0.04%	0.01%	0.07%
B	0.00%	0.00%	0.01%	0.03%	0.53%	98.32%	0.75%	0.06%	0.30%
Caa	0.00%	0.00%	0.00%	0.01%	0.02%	0.79%	97.94%	0.41%	0.82%
Ca-C	0.00%	0.00%	0.01%	0.00%	0.08%	0.32%	1.42%	94.55%	3.62%
Default	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	100.00%

根据表1计算出的一个月期信用评级迁移矩阵

信用评级冲量

假设在一定时间段内信用评级的变化独立于其他时间段，其实并不符合实际。

例如：如果一家公司最近被降级，那么它在接下来一段时间再次被降级的可能性很高。这种现象被称为评级冲量（Ratings Momentum）。

但是对于大部分情况，独立性假设是基本成立的。



二、Vasicek模型

Vasicek模型

《巴塞尔协议II》对银行账户的信用风险资本金要求是基于Vasicek的**高斯Copula模型**的。该方法用于计算贷款组合违约率分布的第X个分位数。



Vasicek

- Oldrich Alfons Vasicek（1942年— ）：捷克裔美国数学家、金融经济学家，布拉格捷克理工大学数学博士，曾在Wells Fargo（1970s）、高盛、KMV Corporation及多家资管公司任职。Vasicek 在金融工程领域有两座里程碑，均以他的名字命名：
- （1）**Vasicek 利率模型**（1977）"An Equilibrium Characterization of the Term Structure"
- （2）**Vasicek 信用组合模型**（1987, 2002）
- 1987 年工作论文："Probability of Loss on Loan Portfolio"（KMV 公司内部报告）
- 2002 年正式发表："Loan Portfolio Value"（《风险》杂志, 2002年12月）
- 地位：现代信用组合风险理论的基石，直接催生了 Basel II/III 的 IRB 资本计量框架

Vasicek

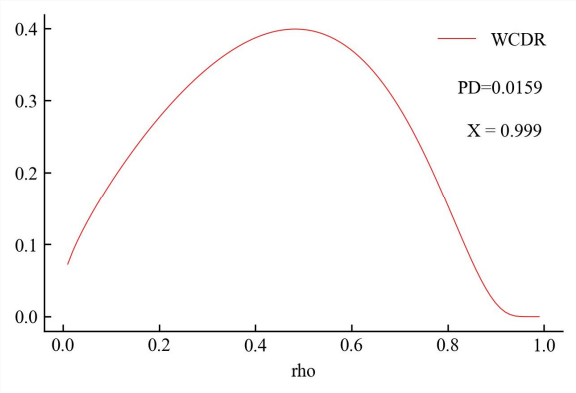
- 2004年，《巴塞尔协议II》正式发布，将 Vasicek 模型纳入内部评级法（IRB），成为全球银行业监管标准。
- 2010至今，《巴塞尔协议III》继续沿用并微调该框架。

定义$WCDR(T,X)$为时长为T的区间内，违约率分布的第X个分位数。

$WCDR$代表最坏违约率（worst case default rate）。

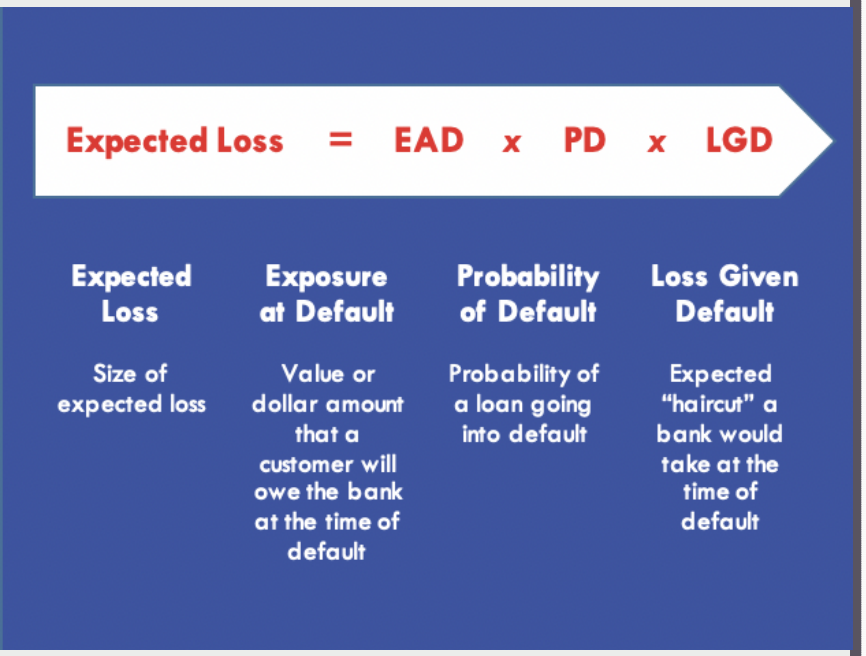
Vasicek模型将$WCDR(T,X)$与违约概率PD以及信用相关性ρ建立联系。

$$WCDR(T,X)=N\left(\frac{N^{-1}(PD)+\sqrt{\rho}N^{-1}(X)}{\sqrt{1-\rho}}\right)$$



Vasicek模型

该模型的缺陷是对尾部相关性的考虑不够充分，采用其他copula代替高斯copula可以对这一点进行修正

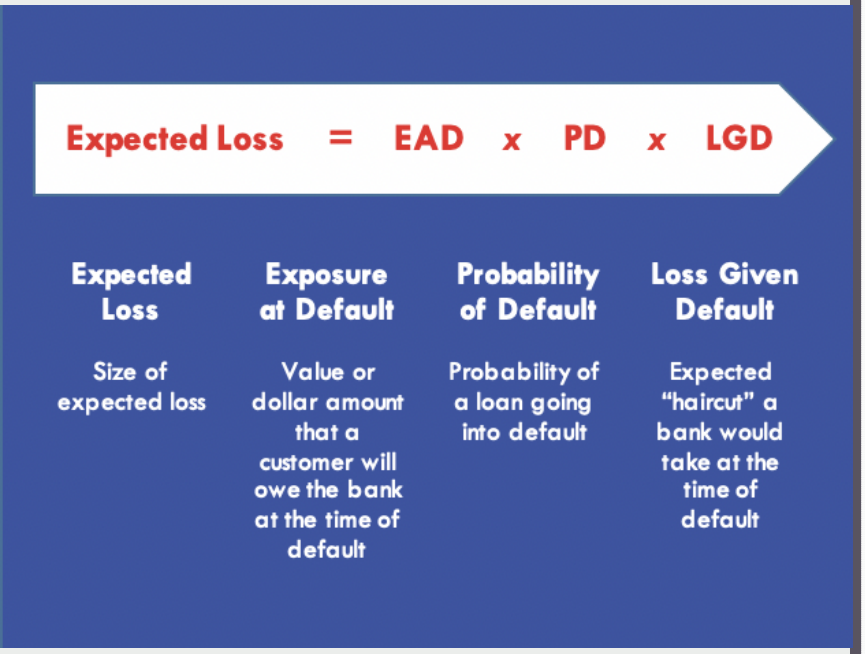


Vasicek模型

对单笔贷款，令EAD代表违约时的风险敞口，

LGD为违约损失，损失分布的第X个分位数

大致为 $WCDR(T,X)\times EAD\times LGD$

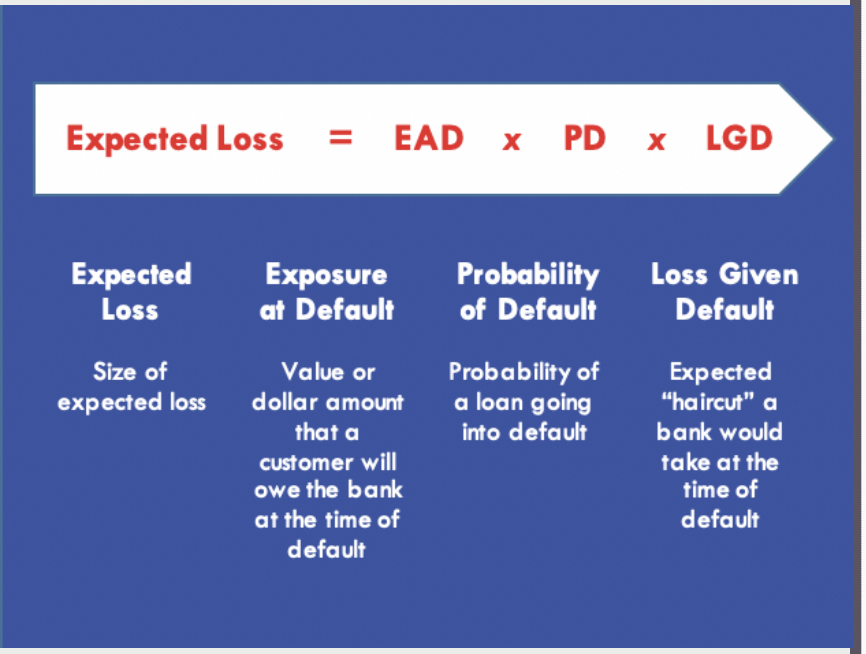


Gordy对Vasicek模型的扩展

一个由n笔贷款组成的贷款组合，其中每笔贷款相比整个组合来说都很小，则损失分布的第X个分位数大致为

$$\sum_{i=1}^{n} WCDR_i(T,X) \times EAD_i \times LGD_i$$

监管资本金等于该式在T=1年，X=0.999时的值。



模型中的参数

有时上式中求和的各项需乘以一个期限调整因子MA_i。

在IRB的基础方法中，银行估计PD的值，EAD和MA的值由《巴塞尔协议II》规定。

在IRB的高级方法中，上述各值全部由银行估计。

用来确定经济资本金时，银行可以自行估计ρ值。

两家公司的ρ应该大致等于它们的资产回报率之间的相关性。

模型中的参数

MA（Maturity Adjustment，期限调整因子）是巴塞尔协议II/III内部评级法（IRB）中用于计算信用风险资本要求的核心参数之一。它反映了一个基本直觉：长期贷款比短期贷款风险更大。

根据巴塞尔框架，MA的计算公式为：

$$MA = \frac{1+(M-2.5)\times b}{1-1.5\times b}$$

其中：

- **M**：贷款的有效期限（Effective Maturity），通常以年为单位，有下限1年、上限5年
- **b**：与违约概率（PD）相关的参数，计算公式为：

$$b = [0.11852 - 0.05478 \times \ln(PD)]^2$$ OSFI

不同 PD 水平下的 WCDR 曲线族

PD 水平	ρ（自动校准）	WCDR 特征
0.5%（投资级）	~0.21	ρ 高，WCDR 对 ρ 敏感，低 PD 被相关性"放大"明显
1.59%（图片原例）	~0.17	中等水平
5%（垃圾级）	~0.13	ρ 较低，曲线较平缓
20%（ distressed）	~0.12	ρ 最低，接近 ρ 下限，WCDR 快速逼近 1

核心规律：PD 越低，Basel 赋予的 ρ 越高，导致 WCDR 被系统性风险放大得更厉害。这反映了"安全资产在危机中同时下跌"的尾部相关性。

巴赛尔协议III对信用风险管理的要求

Categorization of exposures

- Whole sale
- Retail
- Securitization
- Equity

Rating grade assignment/ Segmentation

- **Whole sale**: Assignment of obligor and loss severity rating grades
- **Retail**: Segmentation

Assignment of risk parameters

- **Wholesale:** PD, LGD, EAD & M
- **Retail**: PD, LGD, EAD

Capital requirement

- K x EAD

$$K = LGD \times N\left[\frac{N^{-1}(PD) + \sqrt{R}N^{-1}(0.999)}{\sqrt{1-R}}\right] - LGD \times PD$$

Risk weighted assets

- 12.5 x ∑Capital requirements across non defaulted wholesale & retail exposures



三、Credit Risk Plus

Credit Risk Plus

1997年，瑞士信贷（Credit Suisse）金融产品小组开发了一个计算信用VaR的模型，Credit Risk Plus。该模型使用了保险行业中一种被广泛采用的解析近似法。



Credit Risk Plus

当q的值很小而n的值很大时，我们可以用**泊松分布**来近似二项分布

$$\text{Prob}(m \text{ defaults}) = \frac{e^{-qn}(qn)^m}{m!}$$

即使各贷款之间的违约概率不相等，上式仍然基本成立，只要所有贷款的违约概率都很小，且q等于未来一年中贷款组合中贷款的平均违约率。



Credit Risk Plus

但是我们很难确定未来一年的违约概率q，不同年份之间违约概率可能相去甚远，一个比较方便的假设是违约数量qn服从一个gamma分布，分布的均值为μ，标准差为σ。这时泊松分布变成了一个负二项分布。

$$\text{Prob}(m \text{ defaults}) = p^m (1-p)^\alpha \frac{\Gamma(m+\alpha)}{\Gamma(m+1)\Gamma(\alpha)}$$

其中$\alpha = \mu^2 / \sigma^2$，$p = \sigma^2 / (\mu + \sigma^2)$，

$\Gamma(x)$为*gamma*函数。



蒙特卡洛模拟

在实际中，违约损失具有不确定性，风险敞口也有很多类别，各类别的违约概率各不相同，一种更灵活的方法是蒙特卡洛模拟。步骤如下：

（1）在整体违约率中抽样；

（2）建立各类别违约率与整体违约率的回归模型；

（3）对各类别违约事件进行抽样；

（4）对各类别每一违约事件的损失进行抽样；

（5）计算所有违约的整体损失；

（6）重复以上各步，求得整体损失的概率分布；

（7）由整体损失的概率分布来计算VaR。

1970-2016年所有被评级公司的年百分比违约率

Year	Default Rate	Year	Default Rate	Year	Default Rate
1970	2.631	1986	1.830	2002	2.924
1971	0.286	1987	1.423	2003	1.828
1972	0.453	1988	1.393	2004	0.834
1973	0.456	1989	2.226	2005	0.647
1974	0.275	1990	3.572	2006	0.593
1975	0.361	1991	2.803	2007	0.349
1976	0.176	1992	1.337	2008	2.507
1977	0.354	1993	0.899	2009	4.996
1978	0.354	1994	0.651	2010	1.232
1979	0.088	1995	0.899	2011	0.906
1980	0.344	1996	0.506	2012	1.230
1981	0.162	1997	0.616	2013	1.232
1982	1.040	1998	1.137	2014	0.939
1983	0.900	1999	2.123	2015	1.732
1984	0.869	2000	2.455	2016	2.149
1985	0.952	2001	3.679		

SOURCE: Moody's.

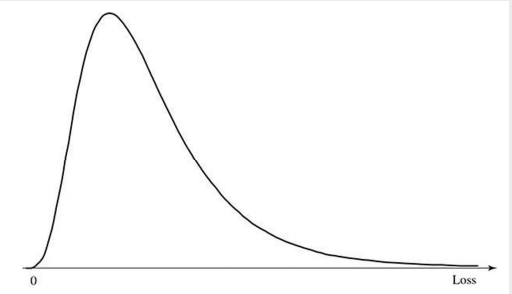


Figure 21.1 General Shape of Probability Distribution of Default Losses

Year	Aaa	Aa	A	Baa	Ba	B	Caa-C	IG	SG	All
1983	0.00%	0.00%	0.00%	0.00%	0.55%	1.05%	21.30%	0.00%	2.02%	0.45%
1984	0.00%	0.00%	0.00%	0.31%	0.26%	2.61%	9.20%	0.09%	1.56%	0.43%
1985	0.00%	0.00%	0.00%	0.00%	0.35%	2.83%	2.49%	0.00%	1.48%	0.37%
1986	0.00%	0.00%	0.00%	0.43%	1.20%	5.20%	8.63%	0.10%	3.09%	0.91%
1987	0.00%	0.00%	0.00%	0.00%	1.10%	1.94%	3.95%	0.00%	1.56%	0.51%
1988	0.00%	0.00%	0.00%	0.00%	0.74%	3.07%	7.21%	0.00%	2.05%	0.73%
1989	0.00%	0.28%	0.00%	0.29%	1.68%	4.25%	12.06%	0.14%	3.34%	1.25%
1990	0.00%	0.00%	0.00%	0.16%	2.18%	8.46%	29.39%	0.04%	6.47%	2.17%
1991	0.00%	0.00%	0.00%	0.16%	2.43%	8.32%	11.56%	0.04%	5.79%	1.77%
1992	0.00%	0.00%	0.00%	0.00%	0.17%	3.88%	8.64%	0.00%	2.52%	0.68%
1993	0.00%	0.00%	0.00%	0.00%	0.39%	2.73%	8.44%	0.00%	2.06%	0.54%
1994	0.00%	0.00%	0.00%	0.00%	0.00%	1.93%	2.01%	0.00%	1.03%	0.28%
1995	0.00%	0.00%	0.00%	0.00%	0.14%	2.00%	6.34%	0.00%	1.61%	0.47%
1996	0.00%	0.00%	0.00%	0.00%	0.00%	0.56%	3.68%	0.00%	0.62%	0.19%
1997	0.00%	0.00%	0.00%	0.00%	0.08%	0.88%	4.11%	0.00%	0.84%	0.27%
1998	0.00%	0.00%	0.00%	0.06%	0.46%	2.17%	5.83%	0.02%	1.81%	0.67%
1999	0.00%	0.00%	0.00%	0.06%	0.86%	3.12%	9.52%	0.02%	3.31%	1.30%
2000	0.00%	0.00%	0.00%	0.26%	1.09%	4.27%	13.93%	0.09%	4.69%	1.87%
2001	0.00%	0.00%	0.12%	0.14%	0.92%	6.88%	23.15%	0.10%	7.38%	2.78%
2002	0.00%	0.00%	0.11%	0.59%	1.24%	3.22%	19.42%	0.26%	5.59%	2.08%
2003	0.00%	0.00%	0.00%	0.00%	0.52%	1.49%	12.44%	0.00%	3.14%	1.08%
2004	0.00%	0.00%	0.00%	0.00%	0.18%	0.33%	5.28%	0.00%	1.10%	0.38%
2005	0.00%	0.00%	0.00%	0.07%	0.00%	0.32%	3.30%	0.03%	0.78%	0.29%
2006	0.00%	0.00%	0.00%	0.00%	0.09%	0.49%	2.50%	0.00%	0.75%	0.26%
2007	0.00%	0.00%	0.00%	0.00%	0.00%	0.05%	2.20%	0.00%	0.46%	0.17%
2008	0.00%	0.33%	0.27%	0.67%	1.55%	2.69%	7.08%	0.41%	3.63%	1.66%
2009	0.00%	0.00%	0.15%	0.58%	1.12%	4.31%	16.55%	0.27%	7.63%	3.13%
2010	0.00%	0.00%	0.09%	0.04%	0.00%	0.20%	4.52%	0.05%	1.61%	0.65%
2011	0.00%	0.11%	0.00%	0.20%	0.09%	0.19%	3.41%	0.10%	1.15%	0.52%
2012	0.00%	0.00%	0.00%	0.04%	0.08%	0.31%	4.59%	0.02%	1.61%	0.71%
2013	0.00%	0.00%	0.05%	0.07%	0.33%	0.55%	3.44%	0.05%	1.47%	0.69%
2014	0.00%	0.00%	0.05%	0.03%	0.08%	0.27%	2.52%	0.03%	1.08%	0.52%
2015	0.00%	0.00%	0.00%	0.00%	0.19%	1.51%	4.14%	0.00%	2.31%	1.09%
2016	0.00%	0.00%	0.00%	0.00%	0.10%	1.14%	6.10%	0.00%	3.10%	1.47%
2017	0.00%	0.00%	0.00%	0.00%	0.24%	0.21%	3.44%	0.00%	1.64%	0.78%
2018	0.00%	0.00%	0.00%	0.00%	0.00%	0.36%	2.71%	0.00%	1.29%	0.62%
2019	0.00%	0.00%	0.00%	0.07%	0.00%	0.83%	4.31%	0.04%	2.16%	1.03%
2020	0.00%	0.00%	0.00%	0.07%	0.09%	2.60%	8.18%	0.04%	4.49%	2.10%
2021	0.00%	0.00%	0.00%	0.00%	0.00%	0.70%	1.48%	0.00%	0.90%	0.41%
2022	0.00%	0.00%	0.00%	0.63%	2.86%	3.58%	2.63%	0.33%	2.95%	1.57%
2023	0.00%	0.00%	0.10%	0.07%	0.49%	0.50%	6.26%	0.08%	3.36%	1.59%
2024	0.00%	0.00%	0.00%	0.00%	0.08%	0.34%	4.98%	0.00%	2.51%	1.13%
2025	0.00%	0.00%	0.00%	0.00%	0.00%	0.46%	5.55%	0.00%	2.78%	1.23%

	Recoveries					Observation Counts				
	Loan	Bond				Loan	Bond			
Industry groups	1st Lien	1st Lien	Sr. Unsec.	Sr. Sub.	Sub.	1st Lien	1st Lien	Sr. Unsec.	Sr. Sub.	Sub.
Aerospace & Defense	67.5%	84.8%	45.5%	23.3%	40.8%	8	2	11	9	7
Automotive	67.2%	48.8%	46.3%	23.9%	49.5%	46	5	39	28	5
Banking		7.0%	41.2%	20.0%	30.3%	0	1	60	7	69
Beverage, Food, & Tobacco	72.2%	55.5%	40.8%	27.5%	34.0%	34	16	34	19	10
Capital Equipment	63.2%	62.6%	34.7%	34.2%	39.6%	38	12	32	30	15
Chemicals, Plastics, & Rubber	66.7%	55.9%	30.8%	23.6%	63.6%	25	24	30	12	2
Construction & Building	58.7%	64.7%	35.7%	25.3%	28.5%	30	12	115	24	20
Consumer goods: Durable	60.9%	62.0%	36.3%	29.3%	29.4%	22	9	13	14	10
Consumer goods: Non-durable	61.9%	48.4%	33.3%	32.9%	38.7%	49	8	31	38	11
Containers, Packaging, & Glass	69.5%	66.4%	52.4%	36.5%	31.9%	13	10	14	11	4
Energy: Electricity	82.4%	82.0%	55.4%	78.9%	22.3%	7	14	30	2	4
Energy: Oil & Gas	68.3%	55.7%	36.9%	38.0%	39.1%	65	46	179	28	36
Environmental Industries	57.2%	13.0%	29.7%	27.1%	13.6%	8	1	7	6	3
Finance**	69.2%	53.4%	44.4%	36.8%	28.9%	15	11	44	11	15
Forest Products & Paper	74.8%	54.4%	37.1%	20.5%	1.5%	14	17	37	9	3
Healthcare & Pharmaceuticals	65.5%	76.3%	37.0%	33.0%	35.3%	67	11	38	30	30
High Tech Industries	55.7%	57.5%	36.9%	26.1%	23.2%	47	20	30	22	38
Hotel, Gaming, & Leisure	69.4%	58.8%	39.3%	32.3%	26.1%	50	51	47	37	18
Insurance	86.2%	96.3%	37.9%	46.0%	43.2%	2	1	10	4	6
Media: Advertising, Printing & Publishing	58.8%	51.4%	28.5%	22.0%	50.9%	38	22	20	23	4
Media: Broadcasting & Subscription	68.8%	47.4%	43.1%	39.1%	29.1%	29	19	55	29	13
Media: Diversified & Production	65.6%	20.8%	36.4%	32.9%	45.2%	6	3	8	3	3
Metals & Mining	57.7%	41.4%	35.5%	32.1%	45.8%	19	39	77	10	11
REIT	67.8%	60.2%	48.4%	13.0%	27.0%	6	3	25	3	7
Retail	56.1%	57.5%	35.0%	34.4%	21.4%	84	46	90	46	28
Services: Business	66.1%	64.0%	44.8%	28.9%	41.5%	76	29	33	11	6
Services: Consumer	61.9%	33.2%	57.8%	49.1%		20	3	8	5	0
Sovereign & Public Finance		31.4%	50.8%			0	3	7	0	0
Telecommunications	60.7%	56.7%	26.7%	31.0%	29.5%	54	31	154	20	19
Transportation: Cargo	66.5%	51.6%	42.0%	35.4%	32.5%	22	19	36	12	5
Transportation: Consumer	84.0%	51.6%	34.0%	43.6%	28.6%	4	14	36	3	14
Utilities: Electric	100.8%	60.6%	54.1%		27.1%	1	11	23	0	2
Utilities: Oil & Gas		85.0%	72.6%		57.5%	0	1	3	0	1
Utilities: Water		40.3%	82.1%		17.8%	0	2	2	0	1
Wholesale	63.8%	53.8%	37.9%	14.2%	37.0%	19	4	15	9	5

2026-10-23 西北农林科技大学陈妍

Annual default rates by broad industry group, 1970-2025

Year	Aerospace & Defense	Automotive	Banking	Beverage, Food, & Tobacco	Capital Equipment	Chemicals, Plastics, & Rubber	Construction & Building	Consumer goods: Durable	Consumer goods: Non-durable	Containers, Packaging, & Glass	Energy: Electricity	Energy: Oil & Gas	Environmental Industries	Finance*	Forest Products & Paper	Healthcare & Pharmaceuticals	High Tech Industries
1970	0.00%	0.00%		0.00%	0.00%	0.00%	0.00%		0.00%			0.00%			0.00%	0.00%	3.70%
1971	0.00%	0.00%		0.00%	0.00%	0.00%	0.00%		0.00%			0.00%			0.00%	0.00%	0.00%
1972	0.00%	0.00%		0.00%	1.09%	0.00%	0.00%	0.00%	0.00%			0.00%			0.00%	0.00%	0.00%
1973	0.00%	0.00%		0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			0.00%			0.00%	0.00%	0.00%
1974	0.00%	0.00%		0.00%	1.11%	0.00%	0.00%	0.00%	0.00%			0.00%			0.00%	0.00%	0.00%
1975	0.00%	3.03%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			0.00%		0.00%	0.00%	4.76%	0.00%
1976	0.00%	0.00%	0.00%	0.00%	1.03%	0.00%	0.00%	0.00%	2.63%			0.00%		0.00%	0.00%	0.00%	0.00%
1977	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.70%			0.00%		0.00%	0.00%	0.00%	0.00%
1978	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.70%			1.32%		0.00%	0.00%	0.00%	3.13%
1979	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%			0.00%		0.00%	0.00%	0.00%	2.78%
1980	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	3.23%	0.00%	0.00%			0.00%		0.00%	0.00%	0.00%	2.70%
1981	0.00%	0.00%	0.00%	0.00%	1.19%	0.00%	0.00%	0.00%	0.00%			0.00%		0.00%	0.00%	0.00%	0.00%
1982	0.00%	2.86%	0.00%	0.00%	0.00%	0.00%	3.33%	0.00%	0.00%			0.97%		0.00%	0.00%	0.00%	4.00%
1983	0.00%	7.69%	0.00%	1.45%	0.00%	0.00%	0.00%	0.00%	0.00%		0.00%	3.19%		0.00%	0.00%	0.00%	1.75%
1984	0.00%	2.78%	0.00%	1.43%	0.00%	0.00%	0.00%	0.00%	0.00%		0.00%	3.98%		0.00%	0.00%	2.22%	4.84%
1985	0.00%	0.00%	0.00%	0.00%	1.76%	2.08%	0.00%	0.00%	1.69%		0.00%	3.41%		0.00%	0.00%	0.00%	1.52%
1986	2.70%	0.00%	0.00%	1.19%	0.00%	1.89%	1.69%	5.88%	0.00%		0.00%	8.31%		0.00%	0.00%	1.43%	0.00%
1987	0.00%	1.75%	0.28%	0.00%	2.45%	0.00%	4.50%	0.00%	0.00%	0.00%	0.00%	5.57%		0.00%	0.00%	3.74%	1.11%
1988	2.17%	0.00%	1.45%	0.00%	0.00%	1.67%	2.70%	4.35%	3.64%	0.00%	4.76%	2.14%	0.00%	0.00%	0.00%	3.54%	1.68%
1989	0.00%	7.41%	1.47%	2.20%	0.00%	0.00%	8.77%	0.00%	3.80%	10.00%	0.00%	0.00%	0.00%	4.63%	0.00%	2.52%	1.76%
1990	0.00%	3.89%	1.81%	5.49%	2.64%	0.00%	16.93%	4.76%	9.96%	0.00%	8.33%	1.48%	0.00%	0.00%	0.00%	1.15%	5.68%
1991	2.04%	4.19%	1.38%	0.00%	4.58%	0.00%	7.69%	5.88%	6.09%	0.00%	4.35%	3.07%	0.00%	0.00%	5.41%	4.90%	2.22%
1992	0.00%	0.00%	0.32%	1.20%	3.03%	0.00%	0.00%	0.00%	5.98%	7.69%	0.00%	0.79%	0.00%	0.00%	0.00%	1.45%	3.35%
1993	2.08%	0.00%	0.30%	1.11%	5.03%	0.00%	1.69%	0.00%	2.00%	0.00%	0.00%	1.50%	0.00%	0.00%	0.00%	0.00%	0.00%
1994	0.00%	0.00%	0.00%	1.03%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	4.00%	0.00%	0.00%	0.00%	2.17%	1.02%	3.03%
1995	0.00%	0.00%	0.00%	1.89%	0.84%	0.00%	5.07%	0.00%	7.63%	0.00%	0.00%	0.63%	0.00%	1.96%	0.00%	0.97%	1.89%
1996	0.00%	0.00%	0.00%	0.82%	0.73%	0.00%	1.05%	0.00%	4.37%	5.56%	0.00%	0.57%	5.00%	0.00%	0.00%	0.00%	1.04%
1997	0.00%	1.35%	0.00%	0.76%	0.00%	0.00%	1.11%	0.00%	6.46%	4.35%	0.00%	0.00%	0.00%	0.00%	0.00%	0.85%	1.82%
1998	0.00%	1.09%	0.30%	0.68%	0.00%	1.59%	0.00%	0.00%	4.42%	0.00%	0.00%	1.28%	5.56%	1.92%	1.18%	3.00%	0.74%
1999	0.00%	0.92%	0.14%	1.82%	1.40%	1.50%	6.42%	10.99%	4.29%	0.00%	0.00%	6.13%	4.35%	1.81%	1.06%	5.22%	2.71%
2000	1.49%	5.07%	0.00%	2.82%	4.21%	1.46%	1.96%	3.57%	14.34%	4.55%	0.00%	0.40%	15.79%	0.61%	7.01%	4.64%	3.21%

Annual default rates by broad industry group, 1970-2025

Year	Hotel, Gaming, & Leisure	Insurance	Media: Advertising, Printing & Publishing	Media: Broadcasting & Subscription	Media: Diversified & Production	Metals & Mining	REIT	Retail	Services: Business	Services: Consumer	Sovereign & Public Finance	Telecommunications	Transportation: Cargo	Transportation: Consumer	Utilities: Electric	Utilities: Oil & Gas	Utilities: Water	Wholesale
2001	8.69%	0.00%	3.17%	3.94%	3.33%	14.09%	0.00%	6.58%	4.20%	3.45%	3.33%	10.08%	4.07%	1.69%	0.73%	0.00%	0.00%	12.68%
2002	1.82%	0.43%	3.45%	14.46%	7.85%	7.56%	0.88%	2.78%	2.13%	3.33%	0.00%	14.46%	4.65%	8.51%	0.00%	1.23%	3.45%	8.62%
2003	2.91%	0.78%	1.67%	5.69%	0.00%	4.53%	0.00%	2.90%	1.10%	0.00%	0.00%	8.04%	3.74%	3.42%	0.32%	0.00%	0.00%	9.33%
2004	4.06%	0.36%	0.00%	4.06%	0.00%	2.00%	0.00%	0.71%	2.26%	0.00%	0.00%	1.49%	0.00%	3.28%	0.30%	0.00%	0.00%	2.78%
2005	0.95%	0.00%	0.00%	1.11%	0.00%	0.90%	0.00%	1.58%	0.00%	0.00%	0.00%	0.42%	0.00%	7.81%	0.30%	0.00%	0.00%	2.86%
2006	0.83%	0.67%	0.00%	2.22%	0.00%	0.00%	0.00%	0.74%	0.00%	0.00%	0.00%	0.47%	3.67%	0.00%	0.00%	0.00%	0.00%	2.44%
2007	0.78%	0.00%	3.15%	0.00%	0.00%	0.00%	0.00%	2.25%	0.74%	2.56%	0.00%	0.89%	1.16%	0.00%	0.00%	0.00%	0.00%	0.00%
2008	9.93%	0.00%	7.37%	2.12%	6.90%	0.81%	2.42%	3.60%	0.64%	0.00%	0.00%	2.37%	5.04%	0.00%	0.00%	0.00%	0.00%	0.00%
2009	17.02%	0.32%	35.06%	23.35%	0.00%	10.31%	5.80%	3.72%	3.13%	7.26%	0.00%	6.27%	7.37%	3.10%	0.29%	0.00%	0.00%	10.26%
2010	4.38%	0.37%	14.25%	3.31%	0.00%	0.00%	0.00%	4.06%	1.81%	0.00%	0.00%	1.53%	1.47%	0.00%	0.00%	0.99%	0.00%	2.56%
2011	3.41%	0.37%	6.38%	0.00%	0.00%	0.00%	0.81%	3.26%	1.24%	1.89%	0.00%	0.00%	5.95%	5.35%	0.00%	0.00%	0.00%	0.00%
2012	3.80%	0.37%	15.22%	1.09%	0.00%	2.20%	0.00%	2.14%	1.62%	6.95%	0.00%	1.68%	1.16%	1.85%	0.53%	0.00%	0.00%	0.00%
2013	2.31%	0.38%	19.27%	0.94%	0.00%	1.95%	0.68%	1.36%	1.94%	0.00%	0.00%	1.64%	2.22%	0.00%	0.00%	0.00%	0.00%	0.00%
2014	1.49%	0.00%	8.22%	0.00%	0.00%	5.29%	0.00%	2.37%	2.47%	3.39%	1.32%	1.59%	0.99%	0.00%	0.00%	0.00%	0.00%	0.00%
2015	2.22%	0.72%	2.22%	1.09%	3.33%	6.79%	0.00%	2.12%	2.56%	4.68%	0.00%	0.54%	0.00%	0.00%	0.25%	0.00%	0.00%	1.61%
2016	0.69%	0.00%	16.19%	0.00%	0.00%	9.36%	0.00%	3.69%	1.46%	0.00%	0.00%	1.53%	2.07%	1.52%	0.24%	0.00%	0.00%	5.94%
2017	0.00%	0.00%	0.00%	4.51%	2.94%	2.03%	0.00%	7.12%	3.96%	1.39%	0.00%	1.51%	3.05%	0.00%	0.00%	0.00%	0.00%	2.94%
2018	0.00%	0.00%	6.58%	1.14%	0.00%	1.93%	0.00%	7.42%	2.34%	0.00%	0.00%	0.99%	1.98%	0.00%	0.24%	0.00%	0.00%	4.31%
2019	2.06%	0.00%	3.33%	1.19%	0.00%	3.17%	0.00%	5.59%	2.85%	1.49%	1.25%	2.99%	2.85%	0.00%	0.47%	0.00%	0.00%	2.82%
2020	7.31%	0.00%	14.18%	0.00%	0.00%	2.66%	0.88%	12.51%	7.01%	9.88%	0.00%	3.13%	4.94%	4.02%	0.25%	0.00%	0.00%	5.47%
2021	1.19%	0.00%	4.00%	0.00%	5.13%	0.68%	0.43%	1.19%	1.88%	0.00%	0.00%	1.09%	1.03%	0.00%	0.00%	0.00%	0.00%	1.39%
2022	2.18%	0.00%	8.94%	1.35%	2.38%	8.53%	0.87%	2.77%	1.06%	1.02%	0.00%	3.24%	4.01%	3.41%	0.74%	0.82%	2.38%	1.25%
2023	5.13%	0.00%	6.35%	4.27%	5.08%	0.83%	0.48%	4.95%	3.91%	3.38%	0.00%	6.28%	2.07%	2.20%	0.50%	0.00%	0.00%	1.11%
2024	3.60%	0.00%	0.00%	9.05%	0.00%	3.33%	0.96%	4.40%	3.72%	2.33%	0.00%	5.42%	1.05%	2.20%	0.26%	0.00%	2.38%	3.39%
2025	4.97%	0.00%	0.00%	4.52%	0.00%	1.75%	0.48%	5.19%	2.38%	3.56%	0.00%	3.69%	1.02%	1.09%	0.00%	0.00%	2.44%	1.18%

蒙特卡洛模拟

在实际中，违约损失具有不确定性，风险敞口也有很多类别，各类别的违约概率各不相同，一种更灵活的方法是蒙特卡洛模拟。步骤如下：

（1）在整体违约率中抽样；

（2）建立各类别违约率与整体违约率的回归模型；

（3）对各类别违约事件进行抽样；

（4）对各类别每一违约事件的损失进行抽样；

（5）计算所有违约的整体损失；

（6）重复以上各步，求得整体损失的概率分布；

（7）由整体损失的概率分布来计算VaR。

1970-2016年所有被评级公司的年百分比违约率

Year	Default Rate	Year	Default Rate	Year	Default Rate
1970	2.631	1986	1.830	2002	2.924
1971	0.286	1987	1.423	2003	1.828
1972	0.453	1988	1.393	2004	0.834
1973	0.456	1989	2.226	2005	0.647
1974	0.275	1990	3.572	2006	0.593
1975	0.361	1991	2.803	2007	0.349
1976	0.176	1992	1.337	2008	2.507
1977	0.354	1993	0.899	2009	4.996
1978	0.354	1994	0.651	2010	1.232
1979	0.088	1995	0.899	2011	0.906
1980	0.344	1996	0.506	2012	1.230
1981	0.162	1997	0.616	2013	1.232
1982	1.040	1998	1.137	2014	0.939
1983	0.900	1999	2.123	2015	1.732
1984	0.869	2000	2.455	2016	2.149
1985	0.952	2001	3.679		

SOURCE: Moody's.

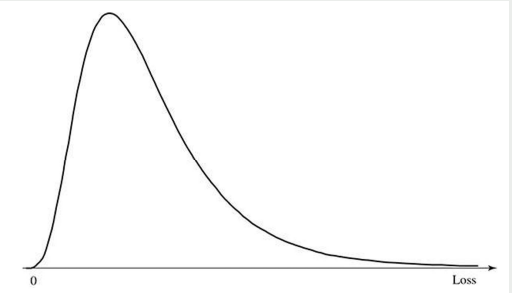


Figure 21.1 General Shape of Probability Distribution of Default Losses



四、CreditMetrics

CreditMetrics

CreditMetrics模型由JP Morgan在1997年提出，基于信用迁移矩阵计算信用VaR，使用高斯copula模型构造信用评级变化的联合概率分布。信用评级可以由内部数据产生，也可以来自专业评级机构。

Vasicek模型和Credit Risk Plus模型仅估算由违约引起的损失概率分布。CreditMetrics模型可以处理降级和违约两种情况。

CreditMetrics

计算一个包含很多对手方的组合1年展望期的信用VaR需要进行蒙特卡罗模拟。

在每一条模拟路径上，确定每个对手方一年后的信用等级，据此计算每个交易对手的信用损失。

如果一年后某个交易对手的信用评级不是违约，则在一年后的时点上对与该交易对手进行的所有交易重新定价，然后求出信用损失。

如果一年后的信用等级是违约，信用损失等于违约时的敞口乘以违约损失率。

计算过程需要用到每个信用等级的新信用价差期限结构。

CreditMetrics

当与某交易对手的交易中包含衍生产品，需要计算信用价值调整量CVA。计算CVA时，需要将从目前开始一直到与该对手进行的期限最长的交易结束时为止的时间段，划分为n个区间。

定义q_i为第i个区间内发生违约的概率；
v_i为区间i 的中点上预期净敞口的贴现值，其中考虑了担保品；
R为回收率。则在当前看来，一年以外违约造成的预期损失为

$$\sum_{i=j}^{n}(1-R)q_i v_i$$

j是自一年后开始的区间。

CreditMetrics

在每次模拟中，如果在第一年没有违约，那么一年时的信用价差期限结构会给出一年以后的n个区间上的违约概率。

假设q_i^*为某次模拟中第i个区间（$i \geq j$）的违约概率。此次模拟中的信用损失为

$$\sum_{i=j}^{n}(1-R)(q_i^*-q_i)v_i$$

如果对手的信用评级在年内得到改善，这一违约损失有可能变成负值。

CreditMetrics

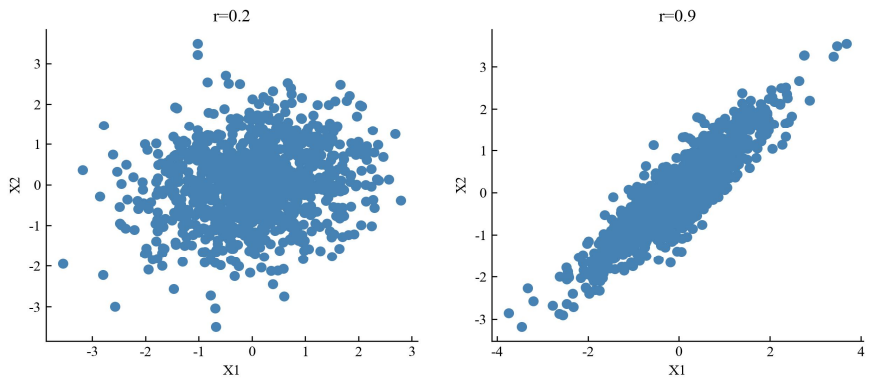
蒙特卡洛模拟的结果会生成一个包括所有交易对手的违约以及降级造成的总损失的概率分布。所需的信用VaR可由这个分布得出来。

相关性模型

假设两家公司股票回报率的相关系数是0.2

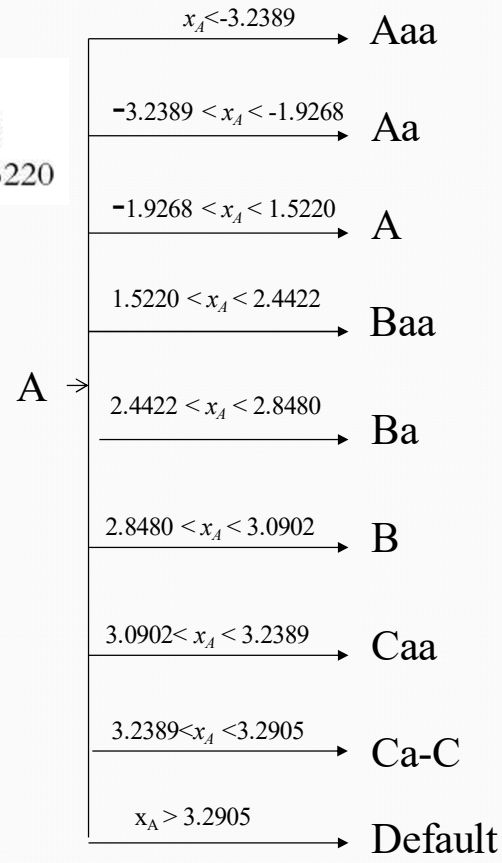
对两个服从标准正态分布的变量x_A和x_B进行抽样，并保证两者的相关系数为0.2

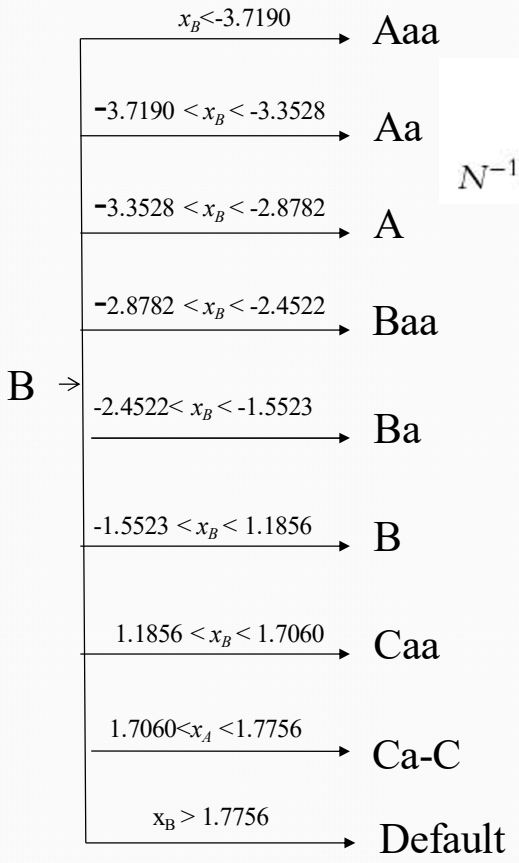
两家公司信用迁移的Copula的相关性一般被设定为等于股票回报的相关性。



$$N^{-1}(0.0006) = -3.2389$$
$$N^{-1}(0.0006 + 0.0264) = -1.9268$$
$$N^{-1}(0.0006 + 0.0256 + 0.9090) = 1.5220$$

评级为A的公司变为评级为Aaa, Aa, A, … 的概率分别为 0.0006, 0.0264, 0.9090, ...





$$N^{-1}(0.0001) = -3.7190$$
$$N^{-1}(0.0001 + 0.0003) = -3.3528$$
$$N^{-1}(0.0001 + 0.0003 + 0.0016) = -2.8782$$

评级为B的公司变为评级为Aaa, Aa, A, … 的概率分别为 0.0001, 0.0003, 0.0016, ...

	Aaa	Aa	A	Baa	Ba	B	Caa	Ca-C	Default
Aaa	90.94	8.36	0.59	0.08	0.02	0.00	0.00	0.00	0.00
Aa	0.87	89.68	8.84	0.45	0.07	0.04	0.02	0.00	0.02
A	0.06	2.64	90.90	5.67	0.51	0.12	0.04	0.01	0.06
Baa	0.04	0.16	4.44	90.16	4.09	0.75	0.17	0.02	0.18
Ba	0.01	0.05	0.47	6.66	83.03	7.90	0.78	0.12	0.99
B	0.01	0.03	0.16	0.51	5.32	82.18	7.39	0.61	3.79

相关性模型

相关性模型

如果一年后某交易对手的信用评级是不违约，那么就要在一年后这个时间点上对与该交易对手进行的所有交易重新定价，然后求出信用损失。

如果一年后的信用等级为违约，信用损失等于违约时的敞口乘以1减去违约回收率（EAD*LGD）

模拟的结果会生成一个包括所有交易对手的违约以及降级造成的总损失的概率分布。

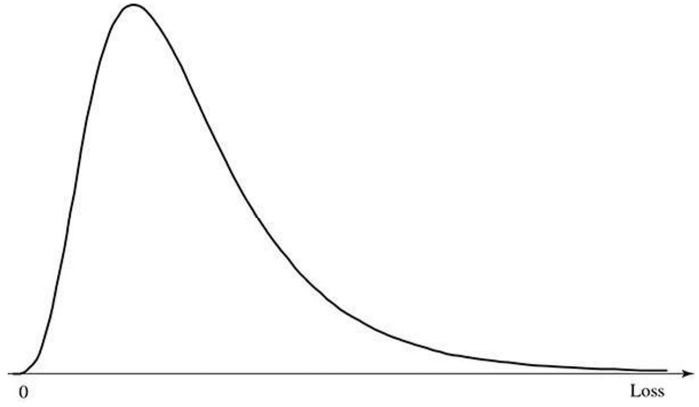


Figure 21.1 General Shape of Probability Distribution of Default Losses